

Nakuja Project

Solid Propulsion Team

One-Grain Static Test — Test Day Report

Document Version: 1.0 (Post-Execution Report)
Reference Procedure: One-Grain Static Test Procedure (attached separately)
Date of Execution: Friday, 10 July 2026
Location: Baseball Pitch (static fire); prior venue (dry run)
Prepared by: Paschal Peters (Data Recorder / Timer)
Dry Run Outcome: **SUCCESS**
Static Fire Outcome: **FAIL** (Ignition Non-Response — Hardware Fault)

1 Purpose

This report documents the events of Friday, 10 July 2026, on which the team conducted a final static test dry run followed by the first live one-grain static fire attempt. This follows the previous dry run report (9 July 2026), in which two procedural faults were identified and resolved. This document records the successful dry run, the static fire test setup and sequence, the fault encountered during the live attempt, and the resulting failure declaration.

2 Part A: Static Test Dry Run

2.1 Overview

Start Time	11:42 am
End Time	11:50 am
Total Duration	8 minutes
Igniter Burn Time	6.034 s (6 s 34 ms)
Procedure Followed	Standard dry run procedure, no deviations
Outcome	Success — no faults observed

The dry run proceeded without incident, confirming ignition circuit continuity, team communication, and timing discipline ahead of the live static fire attempt.

3 Part B: Live Static Fire Attempt

3.1 Overview

Location	Baseball Pitch
Operations Start Time	1:42 pm
Motor Configuration	Fully assembled solid rocket motor with live grain
Procedure Followed	Standard static fire procedure (attached separately)
Total Duration (Start to Declaration of Failure)	1 h 8 min 9 s
Operations End Time	2:50:09 pm
Outcome	Failure — ignition non-response, hardware fault

3.2 Sequence of Events

Pre-Arming Safety Flag: Prior to the first arming attempt, a safety risk was flagged concerning the **load cell sensor**, which was found to be outputting a thrust reading before any force had been applied. This anomaly was investigated and resolved, taking **14 min 21 s** to fix.

Arming and Ignition Attempt: Following resolution of the load cell issue, the ignition sequence was carried out and the motor was armed successfully. However, at the point of firing, the motor did not ignite — no visible or audible response was observed.

Repeated Arm/Disarm Attempts: The team made multiple attempts to arm and disarm the system in an effort to trigger ignition. None were successful.

Stand-Down and Inspection: The motor was disarmed, and the team held a 38-second stand-down period before Ben Muuo approached the test stand to investigate.

Root Cause Identified: Inspection revealed a **short circuit** that caused the MOSFET to fail (fry). As a result, instead of the nichrome wire heating up as intended, the **transmitter wire itself** heated up, which caused the short circuit.

3.3 Fault Summary

Fault	Description	Stage	Resolution Time
#1	Load cell sensor outputting thrust reading before any force applied	Pre-arming safety check	14 min 21 s
#2	Motor armed successfully but failed to ignite; no visible/audible response on first firing attempt	Ignition sequence	N/A — led to Fault #3
#3	Short circuit caused MOSFET failure; transmitter wire heated instead of nichrome wire, triggering the short	Post-arm inspection (root cause)	Declared failure; not resolved on-site

4 Outcome Classification

The static fire attempt is classified as a **FAIL**. The dry run performed earlier in the day validated the ignition circuit, procedure, and team readiness with no issues. However, during the live attempt with the assembled motor and live grain, a hardware-level fault — a short circuit resulting in MOSFET failure and unintended heating of the transmitter wire rather than the nichrome wire — prevented ignition. This is a distinct and more serious class of fault than those observed in the 9 July dry run, as it points to an electrical/wiring failure mode rather than a procedural or battery-related issue.

5 Next Steps

- Full inspection and replacement of the failed MOSFET and affected transmitter wiring prior to any further arming attempts.
- Root cause analysis of the short circuit to determine whether it stems from a wiring/insulation fault, connector issue, or component rating mismatch.
- Verification that the nichrome wire ignition path is electrically isolated from the transmitter line to prevent recurrence.
- Re-inspection of the load cell sensor circuit to confirm the pre-arming zero-offset issue (Fault #1) is fully resolved and not related to the wiring fault.
- Schedule a follow-up dry run to re-validate the ignition circuit before the next live static fire attempt.